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Robotic Process Automation for Patient Data Management in South African Clinics

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1. Background and Context

Providing patient care services accurately, efficiently, and quickly are essential in the healthcare sector. The effectiveness of patient management and healthcare services has substantially increased thanks to Electronic Health Records (EHR) and patient management systems. Yet, because these systems are frequently complicated, controlling them manually can be a difficult undertaking that results in mistakes and delays in patient treatment (Jain,2019). Robotic Process Automation (RPA) has emerged as a promising solution to streamline and automate healthcare processes, including patient data management, appointment scheduling, billing and insurance claims, and medical record management (Aguirre, 2017). RPA is an innovative software technology that facilitates the development, implementation, and control of software robots or bots that imitate human behaviour while interacting with digital systems and software. These software robots can perform tasks such as comprehending on-screen information, executing precise keystrokes, manoeuvring through complex systems, recognising, and extracting data, and executing various pre-defined actions. This technology boasts of quicker and more consistent performance than humans, eliminating the need for breaks or interruptions during operations (Van der Aalst, 2018). RPA software robots can perform repetitive and time-consuming tasks accurately and efficiently, freeing up healthcare professionals to focus on providing quality patient care (Ratia,2018). Hospitals spend 40% of their time on administration, which might be lessened with RPA (Precedence Research, 2022).

Patient data management is a critical process that is carried out in hospitals to ensure effective healthcare delivery. This process involves a series of steps that include data collection, entry, storage, retrieval, analysis, and sharing (Mercury ,2022). The importance of patient data management cannot be overemphasised, as it plays a vital role in various aspects of healthcare, such as decision making, quality improvement, clinical research, public health, and payment reimbursement. Automation has become increasingly prevalent in the healthcare industry, particularly in developed countries (Precedence Research, 2022). These countries have been able to automate hospital processes, including patient data management, due to their technological advancements and the implementation of RPA. Although there has been a greater emphasis on automating critical processes, such as heart monitoring, claims processing, and healthcare supply chain management, developed countries have also been able to automate the processes involved in patient data management (Juell-Skielse, 2022).

RPA in healthcare has been limited in hospitals across South Africa, mainly due to the lack of technological advancements (Cline, 2013). Unlike private hospitals like Netcare and Life Care Healthcare, which have found ways to enhance their healthcare systems technology through the implementation of Electronic Healthcare Records and Electronic Medical Records, there is still a lack of RPA in patient data management processes (Netcare,2022). Although these private hospitals have made strides towards digital transformation, the lack of RPA implementation in patient data management processes still poses a challenge. Patient data management remains a manual and time-consuming process, with healthcare providers having to enter data manually and store it in various formats. Clinics are frequently viewed as a separate entity from hospitals in the world of healthcare facilities, having its own distinct set of operational procedures (Brennan, 2021). However, the majority of the field's current research has focused on technical developments in hospitals, paying relatively little attention to the possibility for similar developments in clinics. As a result, there is a need to refocus

research to explore and advance clinics technical capabilities in order to raise the standard of healthcare delivery as a whole.

2. Problem Statement

Despite having advanced healthcare infrastructure and technology in South Africa, the implementation of RPA for patient data management in the healthcare sector is limited, especially in public hospitals and clinics (Mueller, 2017). Although some private sector hospitals have advanced healthcare technology such as Hospital Information Systems and Electronic Health Systems, the implementation of RPA for patient data management in these systems is yet to be realised (de Villiers, 2020), this indicates that there is still a significant gap in technological advancement in the healthcare sector in South Africa, leading to problems such as manual data entry and increased time spent on paperwork instead of attending to patients. Most research focuses on the implementation of RPA in hospitals, with little mention of clinics, this is due to the focus on large healthcare entities, which include hospitals, as they make up a significant portion of the healthcare industry. Although clinics are healthcare institutions that are dependents of hospitals they should be considered as being hampered by a lack of RPA. Unfortunately, there has been limited research conducted on the implementation of RPA in clinics as the focus is mainly on hospitals. As a result, there is little information available on the adoption of RPA in clinic institutions in South Africa, with most studies centered on major healthcare institutions like hospitals.

3. Purpose Statement

RPA can be implemented in a South African clinic to improve its patient data management processes which will help reduce time spent on paperwork and allow for healthcare workers to have more time to attend to their patients.

4. Preliminary Literature Review

This section provides an overview of the key components that will be explored in the research. It defines the differences between clinics and hospitals, with a specific focus on the patient data management processes. Additionally, it examines the features of RPA tools and the current applications being used in healthcare institutions. The section also delves into the implementation of RPA around the world, including South Africa. This section establishes a comprehensive understanding of the research topic and what will be explored throughout the research.

4.1 Clinics and Hospitals

Clinics and hospitals are both healthcare institutions that offer medical services to patients, but they differ in several ways. The main difference between clinics and hospitals is their size and scope of services. Clinics are usually smaller healthcare facilities that focus on providing

routine preventative care, health screenings, and treatment for minor illnesses or injuries. Hospitals, on the other hand, are larger healthcare institutions that offer a wide range of medical services and can provide care for patients with more severe or complex medical issues (Brennan, 2021). Another significant difference between clinics and hospitals is the level of patient care they provide. Clinics typically offer outpatient care, which means patients do not stay overnight and can receive treatment and go home the same day. Hospitals, on the other hand, provide both outpatient and inpatient care. Patients who require more intensive treatment or surgical procedures may need to stay overnight or for an extended period in the hospital (Helliwell, 1996). In terms of staffing, clinics are often staffed by general practitioners, physician assistants, or nurse practitioners, while hospitals have a more extensive range of medical professionals, including specialists, surgeons, and hospitalists.

The cost of care at hospitals is generally higher than at clinics, due to the specialised services and resources they offer. Clinics may be a more affordable option for routine or preventative care and may be covered by insurance plans with lower copays and deductibles (Hukins, 2001). Clinics are often smaller and located in community settings such as strip malls, retail spaces, or small office buildings. They are generally easily accessible to the local population and often provide services such as primary care, urgent care, or specialty services. Hospitals, on the other hand, are typically larger facilities located in centralised areas within a city or region and are often affiliated with medical schools They may provide a wider range of services than clinics, including inpatient care, emergency care, surgical services, and specialised medical care. Hospitals may also have more advanced medical equipment and technology available to them than clinics (Jiang, 2020).

4.2 Patient Data Management

This section aims to provide an overview of the various processes involved in patient data management within healthcare institutions. As patient data management is a critical aspect of healthcare, it is essential to identify and understand the different stages involved in the process. The following subsections will provide a brief description of each stage of the patient data management process.

Patient data management refers to the collection, storage, organisation, analysis, and utilisation of patient-related information by healthcare institutions to support clinical decision-making, patient care, and population health management. Patient data management is a critical component of healthcare delivery, enabling healthcare providers to deliver high-quality, evidence-based care and improve patient outcomes (Gardner, 1985).

The processes involved in patient data management are:

- Data collection: This involves gathering relevant patient data through various sources, such as medical examinations, laboratory tests, and imaging studies.

- Data entry: The collected patient data is then entered into the EMR or other digital systems using standardised terminologies and codes.

- Data storage: Patient data is stored in a secure and centralised database, allowing authorised healthcare providers to access and retrieve patient information when needed.

- Data retrieval: Healthcare providers can retrieve patient data as needed to support clinical decision-making, patient care, and population health management.

- Data analysis: Patient data can be analysed to identify patterns, trends, and correlations that can inform clinical decision-making, quality improvement, and research.
- Patient data sharing: Sharing patient data is an essential part of patient data management. Healthcare institutions need to ensure that patient data is shared securely and only with authorised individuals. This process can be done through secure email or a secure patient portal.

Both hospitals and clinics carryout patient data management in their institutions to manage patient data.

4.3 Electronic Medical Records Systems

EMR systems are used by healthcare institutions for their patient data management. It is important to understand the definition of EMR systems and how they function before exploring the collaboration between EMRs and RPA.

EMRs are digital versions of traditional paper-based medical records, which contain all the patient's medical and treatment history in one location. EMRs are designed to make patient data management more efficient and accessible. EMRs are used by healthcare institutions to manage patient data, improve patient care, and support healthcare decisions. They are also used to manage billing and administrative tasks, such as scheduling appointments, managing insurance claims, and communicating with patients. EMRs are designed to be used by healthcare providers across different specialties, such as doctors, nurses, and other medical professionals (Kalogriopoulos, 2009). EMRs are typically composed of various modules that can be customised to meet the needs of different medical specialties. Some EMRs are also designed to integrate with other healthcare systems, such as billing and electronic prescribing systems. EMRs offer several benefits to healthcare providers, including improved patient care, better efficiency, and enhanced data accuracy. They also offer improved data security, as patient data is stored in a secure digital format, reducing the risk of data breaches (Wang, 2003).

EMRs have several features that make them a valuable tool in patient data management. One such feature is their ability to store vast amounts of patient data in one centralised location, making it easily accessible to authorised healthcare professionals. This data includes medical histories, diagnoses, test results, medication lists, and other important information that can be used to provide better patient care. Another feature of EMRs is their ability to provide real-time updates to patient records, ensuring that healthcare professionals have the most up-to-date information available. This feature can help reduce the risk of medical errors and improve patient outcomes (Aspden, 2004). EMRs also offer advanced search capabilities, allowing healthcare professionals to quickly and easily search for specific patient information, such as test results or medication dosages. This can save time and improve the accuracy of patient care. In addition, EMRs can facilitate communication and collaboration between healthcare professionals, allowing them to share patient information and work together to develop treatment plans. This which can improve the quality of care

provided to patients and ensure that they receive the best possible treatment (Aspden, 2004).

4.5 Comparison of Clinics and Hospitals in Patient Data Management.

In this section, a comparison between clinics and hospitals in Patient Data Management is presented. The focus is on how these institutions carry out their processes differently with the use of their respective EMR systems. By understanding the differences in data collection, appointment booking, data entry, data storage, data retrieval, data analysis, and data sharing between these two healthcare institutions.

4.5.1 Data Collection

Hospitals tend to collect more comprehensive data on patients due to the nature of their services. This data is collected by filling in a digital form on the EMR system. They collect data such as patient details, vital signs, laboratory results, medication lists, imaging studies, and more detailed medical histories (Abdolkhani, 2019). Clinics, on the other hand, typically collect more basic patient information on their EMR systems such as patient information medical history, vital signs, and reason for visit. The data collected is usually more focused on the specific reason for the patient's visit and the information necessary for the treatment plan. The data collection process in clinics is typically quicker and more streamlined than in hospitals, as the visits tend to be shorter and more routine(Lu, 2010).

4.5. 2 Appointment Bookings

When it comes to appointment bookings for patient data management, there are some differences between hospitals and clinics. In hospitals, appointment booking is usually done through a EMR system that manages all patient appointments for various departments and clinics within the hospital. Patients can usually book appointments through the hospital's website, over the phone, or in person at the hospital's registration desk (Zhang, 2014). On the other hand, clinics may have a more decentralised approach to appointment booking, where each clinic or department manages its own appointments. Patients may need to call each clinic or department separately to book an appointment. Some clinics may also offer online appointment booking through their website or a third-party platform. However, both hospitals and clinics are moving towards more digital solutions for appointment booking, such as online portals and mobile apps, to make it more convenient for patients to book appointments and manage their healthcare (Wang, 2011).

4.5.3 Data Entry

The difference in data entry for patient data management in hospitals and clinics is mainly related to the volume of data and the complexity of the data entry process. In hospitals, patient data entry is usually done by trained medical personnel, such as nurses or medical assistants, who are responsible for collecting and inputting the patient's medical history, diagnosis, and treatment plans. The process of data entry in hospitals is often more complex and time-consuming than in clinics, as hospitals deal with a larger volume of patients and more complex medical cases (Julie ,2019). In clinics, data entry is often done by front desk staff or administrative assistants, who collect basic patient information, such as name, address, and contact details, as well as medical history and insurance information. The process of data entry in clinics is usually simpler and quicker than in hospitals, as clinics deal with a smaller volume of patients and less complex medical cases. The data entry involves EMR systems or even paper-based forms. The data is entered into the system by medical

assistants or other trained staff members, who may use pre-existing templates or forms to capture required information.

4.5.4 Data Storage

In hospitals, patient data is usually stored in a centralised system that allows access to authorised personnel across different departments. This is important because patients in hospitals may require care from various departments and healthcare professionals. They may have more specialised departments, such as intensive care units, emergency rooms, and surgical units, which generate different types of data. As a result, hospitals may require larger storage capacity and more advanced information systems to manage the data (Demigos,2021). Clinics, on the other hand, tend to be smaller and offer more routine care. They may have a more limited set of patient data, consisting mostly of medical histories, test results, and treatment plans. However, the storage of patient data in clinics can still be complex, particularly for clinics that offer multiple types of care or have multiple providers. Due clinic smaller size patients tend to have more consistent healthcare needs that can be managed by a single healthcare professional or department. EMR systems are commonly used in both hospitals and clinics to store patient data in a secure and accessible manner (Atkinson, 2002).

4.5.5 Data Retrieval

Patient data retrieval is an essential aspect of patient data management, and the process involved in retrieving this data may vary between hospitals and clinics. Both hospitals and clinics use EMR systems to store patient data, but the retrieval process may differ. The process for retrieving patient data from an EMR system in a hospital may involve accessing the hospital's network, logging into the EMR system using authorised login credentials, and searching for the patient's information using specific search criteria (McGowan, 2009). On the other hand, in a clinic, the process for retrieving patient data from an EMR system may involve accessing the clinic's network, logging into the EMR system using authorised login credentials, and selecting the patient's name from a list of patients seen by the clinic. The differences in the retrieval process may be attributed to the size and complexity of the hospital's EMR system, which may require more specific search criteria to retrieve data, compared to the clinic's EMR system, which may have a smaller database of patients (Biron, 2014).

4.5.6 Data Analysis

Hospitals typically deal with a larger volume of patients and more complex medical cases than clinics. Therefore, they may require more advanced data analysis tools and techniques to extract insights from the vast amount of data they generate. For instance, hospitals may use data visualisation tools to analyse patterns and trends in patient data, identify risk factors for diseases, and evaluate the effectiveness of medical treatments (CPrime, 2023). Clinics, on the other hand do not require as sophisticated data analysis tools as hospitals, since they typically deal with routine medical cases and a smaller patient population. However, they still need to be able to analyse patient data effectively to provide high-quality care to their patients. Clinics may use data analytics tools to monitor patient outcomes, track medication adherence, and identify gaps in care. In terms of the specific processes involved in patient data analysis, both hospitals and clinics need to follow similar steps on the EMR system. These may include data collection, cleaning, transformation, and analysis. However, the specific tools and techniques used may vary depending on the organisation's size, resources, and goals (CPrime, 2023). While both hospitals and clinics use EMR systems for

patient data management and analysis, the differences in their operations can lead to variations in the way they analyse patient data. Hospitals may require more advanced data analysis tools due to the larger volume and complexity of their patient data, while clinics may focus on more routine analyses. Nonetheless, both types of organisations need to follow similar processes for data analysis on the EMR system.

4.5.7 Data Sharing

When it comes to patient data management, the sharing of data between hospitals and clinics is a critical aspect to ensure continuity of care. hospitals and clinics use EMR systems to store and manage patient data, but there may be differences in how they share this information. In hospitals, patient data sharing may involve multiple departments and healthcare providers, including nurses, physicians, and specialists (Barbazza, 2021). In contrast, clinics may have a smaller team of healthcare providers and less complex data sharing processes. One challenge in sharing patient data between healthcare institutions is ensuring data privacy and security (Barbazza, 2021). Hospitals and clinics must comply with regulations such as HIPAA in the United States, which require that patient data be kept confidential and secure. Therefore, patient data sharing typically involves secure electronic methods, such as encrypted email or secure file transfer protocols. Tools such as Health Information Exchanges (HIEs) and EHR systems are used to facilitate patient data sharing between healthcare institutions (Barbazza, 2021). HIEs are regional networks that allow healthcare providers to share patient data securely across different healthcare organisations. EHR systems are software applications that manage patient data and facilitate communication between healthcare providers. EHRs can also integrate with HIEs to enable broader data sharing. In summary, while hospitals and clinics may have different patient data sharing processes due to differences in team size and complexity, both use EMR systems and tools such as HIEs and EHRs to facilitate secure sharing of patient data (Kalogriopoulos, 2009).

4.6 Robotic Process Automation

This section will focus on discussing the various types of RPA software available and their distinctive features. This discussion will help identify the potential of these software tools in managing patient data in healthcare institutions. Understanding the different types of RPA software and their capabilities will enable healthcare institutions to make informed decisions when selecting an RPA software for their patient data management processes.

RPA is a cutting-edge technology that enables the creation, deployment, and management of software robots that can simulate human actions when interacting with digital systems and software. By automating repetitive and rule-based tasks that were previously performed by humans, RPA has the potential to significantly improve operational efficiency and reduce costs for organisations (Van der Aalst, 2018).

4.6.1 UiPath Studio

UiPath Studio is a powerful visual interface that allows individuals without coding abilities to create robotic procedures. As a modelling tool that uses flowcharts, UiPath Studio simplifies the process of building software robots for automation, making it accessible to a wider range of users (Tripathi, 2018).

UiPath Studio key features involve.

- Drag-and-Drop Workflow Designer: UiPath Studio's intuitive drag-and-drop interface makes it easy to design and create workflows for software robots without the need for programming skills.

- Record and Playback: UiPath Studio enables the recording and playback of user actions, making it easier to automate repetitive tasks without having to create workflows from scratch.

- The object recognition feature uses advanced image and text recognition technologies to identify and interact with user interface elements, such as buttons, text boxes, and drop-down menus.

- Built-in Activities: UiPath Studio comes with a wide range of pre-built activities that can be easily customised to automate various tasks, such as data entry, screen scraping, file and folder manipulation, and email automation.

- Integration with External Applications: UiPath Studio integrates with a wide range of external applications, including Microsoft Office, SAP, and Salesforce, allowing for seamless automation of tasks across different systems.

- Analytics and Reporting: UiPath Studio provides real-time analytics and reporting features, enabling users to track the performance of their software robots and identify areas for improvement.

- Security and Compliance: UiPath Studio is designed with security and compliance in mind, featuring advanced encryption and access controls to ensure the protection of sensitive data.

The capabilities of UiPath Studio make it a potent tool for automating tedious and time-consuming operations, increasing productivity, and cutting expenses for healthcare institutions.

4.6.2 Power Automate

Power Automate is a cloud-based service provided by Microsoft that allows users to create automated workflows and business processes across multiple applications and services. Power Automate enables users to automate repetitive tasks, streamline workflows, and integrate data and processes across different systems, including Microsoft applications like Office 365, Dynamics 365, and Azure, as well as other popular applications such as Salesforce, Dropbox, and Twitter (Critchley, 2020).

Some of the notable features of Power Automate include:

- Workflow Designer: Power Automate offers a visual workflow designer that enables users to create complex workflows by dragging and dropping actions onto a canvas. This feature allows users to create complex workflows without any coding knowledge.

- Connectors: Power Automate comes with a vast library of connectors that allow users to integrate with various applications and services. These connectors include popular services like SharePoint, OneDrive, Dynamics 365, and many more.

- AI Builder: Power Automate includes an AI Builder feature that allows users to create AI models without any coding knowledge. This feature allows users to add AI capabilities to their workflows, such as sentiment analysis and form processing.

- Mobile App: Power Automate offers a mobile app that enables users to access and manage their workflows from anywhere. The app allows users to receive notifications, monitor their workflows, and trigger workflows directly from their mobile devices.

- Templates: Power Automate offers a variety of pre-built templates that allow users to quickly get started with automating common tasks and workflows. These templates cover.

- Analytics: Power Automate offers robust analytics and reporting features that allow users to track and analyse the performance of their workflows. This feature allows users to identify bottlenecks and areas for improvement in their workflows.3632

Power Automate is a strong RPA solution that provides a variety of cutting-edge capabilities for automating healthcare processes and workflows. It is a desirable alternative for healthcare institutions trying to improve efficiency and simplify data management processes due to its simplicity of use, extensive library of connections, and AI capabilities (Critchley, 2020).

4.6.3 Blue Prism

Blue Prism is a leading RPA tool that enables organisations to automate business processes and tasks using software robots. It offers a wide range of features that help businesses streamline their operations, reduce errors, and increase productivity (Chappell, 2017).

One of the key features of Blue Prism is its user-friendly interface that allows even non-technical users to easily build, deploy, and manage software robots. It uses a drag-and-drop approach to automate processes, allowing users to create automation workflows quickly and easily (Chappell, 2017). Another important feature of Blue Prism is its scalability. It can handle many software robots, allowing organisations to automate multiple processes at once. The tool also offers centralised management and monitoring of all software robots, ensuring that they are working efficiently and effectively. Blue Prism also offers advanced analytics and reporting capabilities, allowing users to monitor and analyse the performance of their software robots (Chappell, 2017). This helps businesses identify areas for improvement and optimis e their automation workflows for maximum efficiency. Blue Prism offers a secure platform for automating business processes. It provides robust security features, such as access controls, encryption, and data protection, to ensure that sensitive information is kept safe and secure. Blue Prism is a powerful and versatile RPA tool that offers a wide range of features to help healthcare institutions automate their processes and achieve greater efficiency and productivity (Chappell, 2017). In the context of automating medical processes in healthcare institutions, the RPA tools are well-suited for the task, as they offer features that enable automation of tasks such as patient data management.

4.7 RPA Case studies

In this section, we will explore case studies of RPA implementation in various countries globally, including its application in patient data management and other healthcare-related processes are exploded. The discussion will also include an analysis of the impact of RPA in healthcare in South Africa and the extent of its implementation.

Around the world, RPA has been widely incorporated into healthcare systems, including in hospitals in Sweden and the Netherlands, to reduce errors and boost productivity (Juell-Skielse, 2022). As part of a research study on the application of RPA in healthcare, a Master's student in the Netherlands investigated the areas where RPA was being utilised in hospitals. The research findings indicated that RPA was being implemented in several hospital operations, including remote heart monitoring and supportive functions such as patient data management, finance in health, human resources, and cardiology. The findings suggest that RPA is being utilised in diverse aspects of healthcare operations, in hospitals. These results provide valuable insights for healthcare professionals and decision-makers looking to implement RPA technology in hospitals and other healthcare settings in Netherlands (Juell-Skielse, 2022).

Another investigation into RPA was identified in a study done in Swedish hospitals. The study was conducted to investigate the current state of RPA adoption in Swedish hospitals and the processes that have been automated (Smits, 2023). The study aimed to answer questions surrounding RPA implementation in hospitals and the specific areas where it is being utilised the findings showed that three regions had adopted RPA implementation in their hospitals. Furthermore, it was observed that most of the processes automated using RPA were in the field of finance, representing more than 50% of the identified processes (Smits, 2023). Administration and human resources were the next most frequently automated areas. However, it was noted that only a few administration processes implemented were closely related to healthcare, and these were mainly administrative tasks for patient data management. In one instance, a trial process was implemented in vibration-induced grasp reflex, where an RPA process was used to calculate suggestions for medicines to prescribe in a specific case. These results offer valuable insights into the current state of RPA adoption in Swedish hospitals and the specific areas where RPA is being utilised (Smits, 2023).

London hospitals have been adopting RPA technologies to streamline their processes, with Blue Prism being one of the software tools utilised for this purpose. Within the realm of RPA, Blue Prism stands out as a leading software company that specialises in developing cutting-edge RPA software. The software is designed to automate complex, end-to-end processes for businesses, delivering unparalleled efficiency gains and cost savings. Blue Prism's RPA software has proven to be an industry standard, with its robust and reliable capabilities earning it a well-deserved reputation as a powerful tool for enhancing organisational productivity and effectiveness. The deployment of Blue Prism enabled the hospitals to automate tasks such as appointment booking, ordering, and tracking supplies, and managing patient records, leading to increased efficiency and improved patient outcomes (Sam,2021). Kings College Hospital (KCH) based in London implemented a Blue Prism software for its orders. To streamline KCH operations, KCH deployed a Blue Prism digital workforce to assist with managing the large volume of orders that were being processed daily. Prior to implementing the digital workforce, KCH was logging thousands of orders manually, which resulted in delays and increased turnaround time for goods to be delivered. The Blue Prism digital worker was programmed to automatically contact the supplier if

ordered goods had not been received within the specified time and request an update. Upon receiving a reply, the digital worker would then determine the relevant department and forward the supplier's response for follow-up. This automated system has proven to be highly efficient, reducing the time between order and delivery from eight days to just three. KCH also implemented Blue Prism to automate other patient data management processes, such as appointment bookings, storing patient data, analysis of patient data, resulting in further time and cost savings.
The successful deployment of the Blue Prism digital workforce at KCH showcases the potential benefits of RPA technology in streamlining healthcare operations and improving efficiency.

The implementation of advanced, self-regulatory technologies such as Artificial Intelligence (AI) and machine learning has led to a surge in the adoption of RPA in the healthcare industry, particularly in North America (Precedence Research, 2022). This trend is expected to continue over the projected year, with North America holding the largest market share. The United States government is actively funding and managing campaigns to encourage private healthcare businesses to adopt automated solutions for efficient workflow.

Report Coverage	Details
Market Size in 2022	USD 2.9 Billion
Market Size by 2030	USD 6.2 Billion
Growth Rate from 2022 to 2030	CAGR of 26.01%
Base Year	2021
Forecast Period	2022 to 2030
Segments Covered	• By Solution • By Operations • By Application
Regions Covered	• North America • Europe • Asia-Pacific • Latin America • Middle East & Africa

Figure 1 Robotic Process Automation in Healthcare market shares (Precedence Research, 2022).

Figure 1 depicts the distribution of market share in the field of RPA across various regions worldwide. The study reveals that North America is expected to hold the largest market share in the global RPA market, with the potential to drive the overall growth of the industry. Furthermore, the data highlights the RPA market share of other regions such as Latin America, Asia Pacific, Europe, and the Middle East Africa, indicating the diverse nature of the RPA market and the varying levels of adoption and growth across different regions (Precedence Research, 2022).

In March 2019, the US Federal government approved a budget to promote the implementation of RPA and AI in order to provide better services to the American public. This has resulted in the development of 1,000 RPA robotic automation systems and the

release of 1.5 million hours of capacity, expanding the market as enterprises and government organisations increasingly utilise automation software, processes, and solutions (Precedence Research, 2022).

Over the past few years, various applications have emerged and experienced rapid growth in South Asian countries, drawing the interest of several American and European multinational corporations (MNCs). One such application is the RPA of appointment scheduling and patient data management which eliminates the need for human intervention and makes the process quick and efficient. Customers appreciate receiving timely notifications and reminders of their appointments, which makes them feel valued and important (Precedence Research, 2022).

In 2021, the global market was dominated by the software sector, which is known for its versatility and diverse range of applications, and this trend is expected to continue in the future. The Asia Pacific region has become a major global centre for software and related services due to rapid urbanisation and technological advancements. The consulting services segment currently holds the largest share of market value, primarily due to heightened awareness and demand from clients (Precedence esearch, 2022). This indicates a growing preference for software solutions and services in the market. The preceding statements highlight the prevalence of RPA in the healthcare industry. The evidence indicates that RPA has predominantly been adopted in developed countries and is expected to continue to grow due to funding and technological advancements in these nations. This suggests that the first world will continue to lead the way in the implementation of RPA in healthcare (Precedence Research, 2022).

The implementation of RPA in healthcare is mostly prevalent in developed countries like the United States, Canada, and the United Kingdom, while it has not gained widespread adoption in many developing countries such as South Africa. Despite being one of the most advanced African countries in healthcare infrastructure and technology, the use of RPA in the healthcare sector in South Africa is limited (Cline, 2013). However, private hospitals like Netcare have pioneered the use of an EMR system, which although not RPA implementation, represents an advancement in health technology beyond regular healthcare systems. Many hospitals around the world are yet to adopt the capability of remotely accessing real-time patient clinical information and test results (de Villiers, 2020). Netcare EMR system enables the electronic management and documentation of all patient records and seamless integration with pathology laboratories, radiology, and the blood bank (Netcare,2022). Netcare has also integrated medical equipment and devices in ICUs and theatres using digital technology. They have also developed a mobile platform with an iPad for voice-to-text note dictation and handwriting recognition, as well as computerised physician order entry for electronic ordering of investigations and requests. The results are then electronically returned to CareOn for healthcare workers to access critical clinical information automatically (Moon, 2022).

Life Healthcare, a leading private hospital group in South Africa, has also adopted EHRs to manage patient data. The EHR system, known as Clinicom, offers clinicians access to patients' health information from any location and at any time, as well as tools for clinical decision-making and care coordination. The system facilitates electronic ordering and receiving of laboratory and radiology tests, as well as medications, which minimises errors and increases efficiency. In addition, the system provides real-time alerts and notifications to clinicians for critical results and events, improving patient safety and quality of care (Life Healthcare, 2021). The hospital has been successful in implementing contemporary EHRs, but RPA deployment in the hospital is still an open question especially for patient data management.

Chris Hani Baragwanath Academic Hospital, a public hospital located in Johannesburg, South Africa, has implemented a digital health platform known as the Hospital Information System (HIS). This system comprises modules dedicated to patient data management, data entry, and other clinical workflows, with the goal of improving the efficiency and quality of healthcare services provided (Advertorial ,2016). The existing system operates solely through manual control by users, with no automated processes in place for patient data management.

Mediclinic, a private hospital group in South Africa, has implemented EHR to manage patient data across its facilities. The EHR system, called Meditech, has various modules that manage patient data, enable data entry, and support clinical decision-making. The system's goal is to provide a unified view of patient health information throughout the care continuum, enhancing the quality of healthcare services provided by the hospital group (Mediclinic,2020).

One study found that the implementation of RPA in healthcare in South Africa was hampered by factors such as the lack of awareness and understanding of RPA technology, limited financial resources, and inadequate IT infrastructure in some healthcare facilities (MacQuilkan, 2018). This contrasts with developed countries where RPA has been widely implemented in healthcare and is seen as a key tool for improving efficiency and reducing costs. Although RPA technology seems to be limited in South African healthcare other implementations of modern-day technologies have been made to hospitals especially in the private sector (Netcare,2022). Advanced systems such as EHRs and EMRs have been adopted, while adoption of RPA technology in South African healthcare remains limited, despite this, these hospitals have managed to leverage technology to improve their operations and enhance patient care. However, RPA has not yet been implemented in these systems, indicating that there is still room for further technological advancement in the healthcare sector in South Africa. The apparent dearth of RPA implementation in both public and private healthcare institutions highlight the likelihood that clinics also lack RPA implementation (Cline, 2013). Clinics are viewed as smaller healthcare facilities compared to hospitals, which receive more attention in the literature on RPA implementation. In the case of South African clinics, there is a noticeable scarcity of evidence on the implementation of RPA technology.

To broaden our understanding of the relationship between RPA and health technological advancement in developing countries, a study was conducted, which examined the implementation of healthcare systems in a hospital in Nigeria, a middle-income country. The study found that the majority (95.0%) of people have access to computers and the Internet. A larger percentage of the participants showed inadequate knowledge of healthcare management software (46.3%) and database management (47.8%) (Adeleke,2015). The vast majority have used computer (95.0%) and the Internet (93.6%) for more than 3 years and many (86.2 percent) expressed a desire for further ICT training. It is important to note that this study was focused on the implementation of typical healthcare systems and did not include RPA technology (Adeleke,2015). Therefore, it highlights the need for healthcare facilities in Nigeria and extension Africa to first focus on improving their existing healthcare systems before considering the adoption of advanced technologies such as RPA.

5. Research Objectives

This research study centers on the installation of an RPA software into clinic EHR systems. Subsequently, automated patient management processes will be designed and developed using the RPA software that has been installed in the clinic EHR systems. The finalised processes will then be integrated with the EHR systems to facilitate automation.. This will serve as the project's artifact.

The aim will be to improve existing patient management processes on a clinic management system by implementing RPA software. This study aims to provide recommendations on how RPA can be effectively utilised to enhance patient data management in a clinic. By achieving the research objectives stated below, this study aims to contribute to the improvement of patient care and offer insight for future efforts to advance the healthcare institutions in South Africa.

The research objectives for this study are:

- RO1: To assess the advantages and obstacles that come with implementing RPA in healthcare institutions, specifically in patient data management.

- RO2: To evaluate the extent to which RPA has been adopted in patient data management in healthcare institutions.

- R03: To analyse the current patient data management processes in a selected clinic in South Africa and determine which ones are suitable for the implementation of RPA.

- R04: To investigate suitable RPA technologies and methodologies for the implementation of patient data management automation in South African clinic.

- RO5: To implement RPA technology in a South African clinic) for the management of patient data.

- RO6: To assess the impact of the RPA implementation for patient data management in South African clinic.

6. Design Science Research Methodology

The present research project involves an RPA software to be integrated into a EHR system at a clinic. To achieve this goal, the Design Science Research Methodology (DSRM) has been identified as the most appropriate research approach. The DSRM is a methodology that emphasises the creation of innovative artifact that contribute to the field of study. In accordance with this methodology, a comprehensive literature review will be conducted to gather useful information and guide the development of the RPA software.

The DSRM approach is deemed suitable for this research project because the development of an innovate artifact is a primary objective which will be the implementation of an RPA

software in EHR system for a clinic, with the aim of contributing to the healthcare field. In this regard, the research will follow the DSRM's recommended phases, which will be extensively discussed in the dissertation. These phases include the:

6.1 Awareness of Problem

The Design Science Research Process commences with the identification of the problem to be addressed by the research. In this study, the problem is identified as the lack of RPA implementation in South African clinics to effectively manage patient data. To accurately identify and define the problem, an extensive literature review will be conducted to investigate the current state of RPA implementation in clinic management systems. The literature review aims to produce a proposal for a new research initiative that will focus on implementing RPA software in a clinic to address the identified problem. This initial phase is of utmost importance as it lays the foundation for the subsequent stages of the DSRM methodology. The success of the overall research project depends largely on the effectiveness of this phase, which will ensure that the problem is well-defined and aligns with the research objectives (Rothenber, 2014)

6.2 Suggestion

The second phase of the DSRM methodology involves identifying the patient data management processes that need to be automated in a clinic, following the identification of the problem in the first phase. This process will involve conducting interviews and questionnaires with healthcare workers to gain an understanding of the current patient data managed in the EHR system and the processes involved. The information obtained will then be used to identify the patient data management processes that will require automation. Then after next step will involve identifying an appropriate RPA software that can effectively address the automation of the identified processes. To achieve this, various techniques such as brainstorming, literature review, and expert opinions will be utilised to generate ideas on the best RPA software to use for the automation of the processes. The identified RPA software solutions will be evaluated based on their capability to automate patient data management processes in the clinic. The features, capabilities, and limitations will be analysed. UiPath, Power Automate, and Blue Prism will be investigated in-depth to determine their suitability. The selected RPA software will need to meet specific criteria to be deemed suitable for addressing the identified problem. This phase is critical as it forms the link between the proposal and the design of the implementation idea. The output of this stage is a set of potential solutions that will be evaluated and refined in the subsequent stages of the DSRM (Keuchler, 2015).

6.3 Development

In the development phase of the DSRM methodology, the proposed solution is put into practice through the creation of an artifact. In the context of this study, the chosen patient data management processes and RPA software are implemented and integrated into the existing EHR system at the clinic, to automate the processes. To achieve this, the processes to be automated will be modelled using software tools such as Visio. The implementation of the RPA software onto the EHR system follows, with the installation of the software onto the computers that have the EHR system or external computers that have the system. The

modelled RPA processes are then implemented onto the EHR systems. Afterward, thorough testing is conducted to evaluate the effectiveness of the RPA software in manipulating the EHR system. Iterative development is then carried out in phases until the final processes are done. The output of this will be an artifact that demonstrates the validity of the proposed solution and its potential to address the identified problem (Keuchler,2015).

6.4 Evaluation

Once the RPA software has been integrated and implemented into the clinic's EHR system, an evaluation will be conducted to assess the impact of the RPA software on the EHR system. This evaluation will involve a series of instructions that participants will be asked to follow in order to determine whether the automated processes have been successful in performing their intended tasks. Additionally, the evaluation will aim to identify any errors or problems that may have been missed during the development phase, which can then be addressed and resolved in subsequent phases of the methodology. This evaluation is a critical step in determining the effectiveness of the RPA software in improving patient data management processes in the clinic and will inform any necessary adjustments or improvements that may need to be made before the final product is produced (Rothenber, 2014).

6.5 Conclusion

The conclusion phase of the DSRM methodology involves assessing the effectiveness of the implemented RPA software and the success of the patient data management processes being automated. This will also involve the documentation of the steps taken during the methodology, any challenges encountered, and the outcomes achieved. The conclusion will also include recommendations for further improvements and future research directions for RPA implementation in clinics and other (Keuchler,2015).

6. 7 Iterative Approach

An iterative approach is a fundamental aspect of the Design Science Research (DSR) methodology, which involves a continuous process of designing, developing, evaluating, and refining a solution to a problem. This approach enables the researcher to refine and improve their solutions based on feedback obtained from the evaluation process, which can be used to inform subsequent iterations (Larman, 2004). In line with the DSRM methodology, the implementation of the RPA software involves an iterative process that involves cycles of modelling, integration, and testing of the EHR system's processes. Each process is modelled, integrated, and tested to determine its effectiveness. If a process does not work effectively, necessary modifications are made until the process achieves its intended goal. The iterative process plays a significant role in the DSR methodology as it offers a flexible and adaptable framework for developing a solution to address complex challenges. Through the iterative refinement and upgrading of the solution, an artefact can develop innovative and efficient solutions that can have a significant impact in the field.

7. Scope

The aim of this research is to explore the implementation of RPA in patient data management at a university clinic. This implementation will be conducted to leverage the benefits of RPA, which include reducing mistakes and errors, increasing efficiency, and minimising the time required for mundane tasks. Additionally, this research will evaluate the implementation of RPA by analysing both its benefits and challenge.

8. Time Schedule

Deliverable	Date
Master's Proposal	May 2023
Chapter 1 & 2 Introduction and Awareness	August 2023
Chapter 3 Literature review and Suggestions	October 2013
Chapter 4 Development	February 2024
Chapter 5 Evaluation	June 2024
Chapter 6 Conclusion	August 2024
First Draft	September 2024
Second Draft	October 2024
Final Submission	November 2024

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SIGNOFF

I __Phillip Sibanda_____ (name) hereby declare that this assignment is my own work.

I ___Phillip Sibanda_____ (name) also declare that my supervisor has seen a copy of this assignment and has reviewed it.

Signature: ____*P Sibanda*_____ Date: __09/03/2023_____